

```
%{
#include<stdio.h>
#include<stdlib.h>

int yylex(void);
void yyerror(const char *s);
}%

%token NUM
%left '+' '-'
%left '*' '/'

%%
expr:e {
    printf("Result:%d\n",$$);
    return 0;
}

e:e '+' e { $$ = $1 + $3; }
| e '-' e { $$=$1-$3; }
| e '*' e { $$ = $1 * $3; }
| e '/' e {
    if ($3 == 0) {
        printf("\nError: Division by Zero\n");
        exit(1);
    }
    $$ = $1 / $3;
}
| '(' e ')' { $$ = $2; }
| NUM { $$ = $1;}
;
%%

int main() {
    printf("Enter the arithmetic expression\n");
    yyparse();
    printf("Valid expression\n");
    return 0;
}

void yyerror(const char *s) {
    printf("\nInvalid expression: %s\n", s);
    exit(1);
}
```